

AMRITJOT SINGH

SOFTWARE ENGINEER | FULL STACK | AI/ML & CLOUD

Melbourne, VIC, Australia | 0420 291 207 | aj044223@gmail.com | linkedin.com/in/amritjot-singh4322 | github.com/amrit4322

PROFESSIONAL SUMMARY

Software Engineer and Master of Data Science graduate with experience developing full-stack applications, backend services, APIs, microservices and AI/ML systems. Proficient in Python, Java, C++, JavaScript and TypeScript, with hands-on experience across React, Node.js, FastAPI, databases, Docker and cloud technologies. Experienced in data structures and algorithms, debugging, testing, performance optimisation and building maintainable software systems. GovHack 2025 Victorian State Winner with demonstrated experience improving backend response times, automating workflows and developing ML-enabled applications.

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript, TypeScript, C#, SQL

Computer Science: Data Structures & Algorithms, OOP, Software Design, Problem Solving

Front End: React, Next.js, HTML, CSS, Tailwind CSS

Back End: FastAPI, Node.js, Express, Django, REST APIs, gRPC, Microservices

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI/ML: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLOv8, EfficientDet, Stable-Baselines3

Cloud & DevOps: Azure, AWS, Docker, Kubernetes, Git, GitHub Actions, CI/CD

Engineering: Unit & Integration Testing, API Testing, Performance Testing, Debugging, Code Review, Agile/SDLC

PROFESSIONAL EXPERIENCE

Software Engineer | Full Stack Developer - Antier Solutions Pvt. Ltd. | Jul 2023 - Jun 2024

- Developed full-stack and backend applications using Python, FastAPI, Node.js, React, JavaScript and microservice architectures.
- Designed and implemented REST and gRPC APIs supporting communication between distributed application services.
- Built analytics and dashboard functionality for BlockAgile, reducing manual reporting effort by approximately **50%**.
- Optimised backend and service communication workflows, reducing response times by approximately **30%**.
- Debugged application and API issues, tested service behaviour and collaborated through Git-based development and code-review workflows.
- Developed and evaluated computer-vision functionality using YOLOv8 and EfficientDet, improving object-detection accuracy by approximately **12%**.

Operations & Inventory Coordinator - ID Logistics, Amazon Client Operations | Melbourne, VIC | 2026 - Present

- Use warehouse management systems to investigate inventory and order discrepancies and resolve operational exceptions in a high-volume environment.
- Analyse system and stock information, perform reconciliation and coordinate issue resolution with operational teams.

SELECTED ENGINEERING PROJECTS

CoBuild - GovHack 2025 Victorian State Winner | Node.js, React, Python, APIs, Data Processing

- Led solution architecture and developed React, Node.js and Python components for a data-driven urban-planning platform.
- Built APIs, mapping and scoring workflows that transformed public datasets into actionable planning information.

PIPER - Physical Reasoning in AI | Python, PyTorch, Stable-Baselines3, Reinforcement Learning

- Developed a Python-based experimental benchmark for comparing reinforcement-learning and AI reasoning approaches on physics-based tasks.
- Implemented simulation environments, reward mechanisms and PPO-based reinforcement-learning experiments; analysed performance and failure cases.

Note Wallet - Distributed Backend Services | Python, FastAPI, gRPC, Docker

- Developed containerised backend services using Python, FastAPI and gRPC; improved service response performance by approximately **30%**.
- Tested and debugged APIs and service interactions to improve system reliability.

BlockAgile / ConnectUs | React, Node.js, APIs, Databases, Docker

- Developed full-stack functionality, real-time dashboards and secure APIs; conducted performance and stress testing for systems supporting **100+ concurrent users**.

EDUCATION

Master of Data Science (Professional) - Deakin University, Australia | 2024 - 2026

Research: Physical reasoning and reinforcement learning, comparing learning-based and AI reasoning approaches using Python-based simulation and experimentation.

Bachelor of Engineering - Computer Science & Engineering - Chitkara University, India | 2020 - 2024 | **CGPA: 9.51/10**

Relevant foundations: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks and Machine Learning.

ACHIEVEMENTS & CERTIFICATIONS

- **GovHack 2025 - Victorian State Winner**
- **Microsoft Azure AI Fundamentals (AI-900)**
- IBM AI Developer Professional | Blockchain Specialisation - University at Buffalo
- Technical Speaker - Docker for Founders & Startups, Microsoft Melbourne | 2026